

Windows Registry

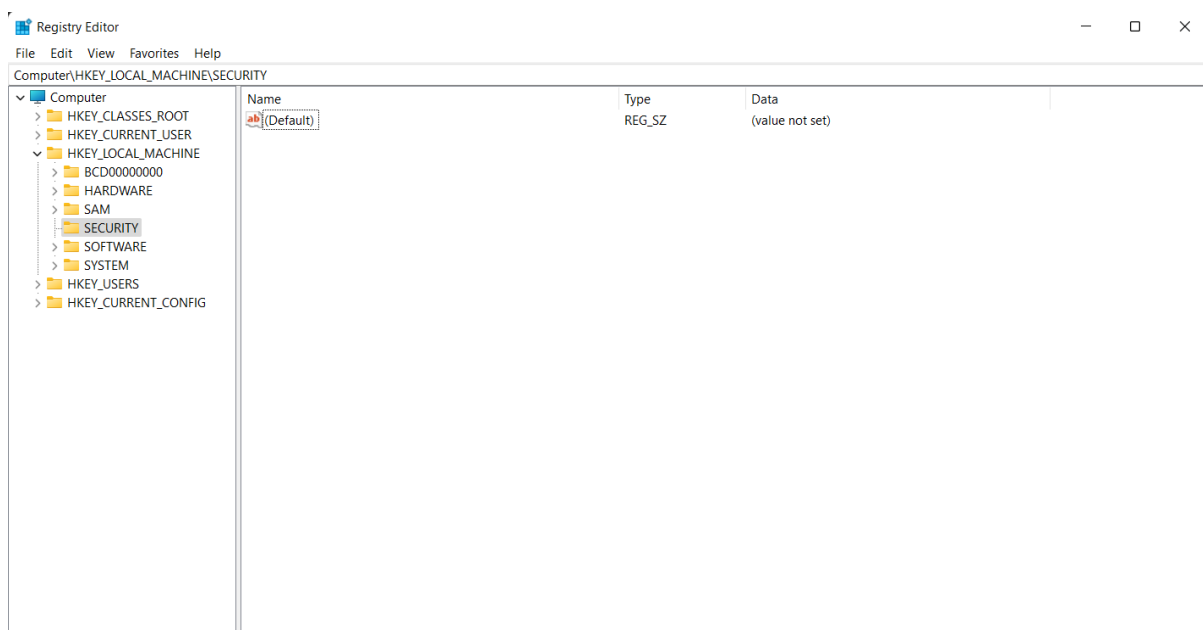
Windows Registry

- a collection of databases that contains the system's configuration data
- This configuration data can be about the hardware, the software, or the user's information.
- It also includes data about the recently used files, programs used, or devices connected to the system.
- The registry on any Windows system contains the following **five root keys**:

Folder/predefined key	Description
HKEY_CURRENT_USER	- Contains the root of the configuration information for the user who is currently logged on. - The user's folders, screen colors, and Control Panel settings are stored here. - This information is associated with the user's profile. - This key is sometimes abbreviated as HKCU .
HKEY_USERS	- Contains all the actively loaded user profiles on the computer. - HKEY_CURRENT_USER is a subkey of HKEY_USERS. - HKEY_USERS is sometimes abbreviated as HKU .
HKEY_LOCAL_MACHINE	- Contains configuration information particular to the computer (for any user). - This key is sometimes abbreviated as HKLM .
HKEY_CLASSES_ROOT	- Is a subkey of <code>HKEY_LOCAL_MACHINE\Software</code> . - The information that is stored here makes sure that the correct program opens when you open a file by using Windows Explorer. - This key is sometimes abbreviated as HKCR . - Starting with Windows 2000, this information is stored under both the HKEY_LOCAL_MACHINE and HKEY_CURRENT_USER keys. 1. The <code>HKEY_LOCAL_MACHINE\Software\Classes</code> key contains default settings that can apply to all users on the local computer. 2. The <code>HKEY_CURRENT_USER\Software\Classes</code> key has settings that override the default settings and apply only to the interactive user . - HKEY_CLASSES_ROOT key provides a view of the registry that merges the information from these two sources. - <code>HKEY_CLASSES_ROOT</code> also provides this merged view for programs that are designed for earlier versions of Windows. - To change

	the settings for the interactive user, changes must be made under <code>HKEY_CURRENT_USER\Software\Classes</code> instead of under <code>HKEY_CLASSES_ROOT</code> . - To change the default settings, changes must be made under <code>HKEY_LOCAL_MACHINE\Software\Classes</code> . - If you write keys to a key under <code>HKEY_CLASSES_ROOT</code> , the system stores the information under <code>HKEY_LOCAL_MACHINE\Software\Classes</code> . - If you write values to a key under <code>HKEY_CLASSES_ROOT</code> , and the key already exists under <code>HKEY_CURRENT_USER\Software\Classes</code> , the system will store the information there instead of under <code>HKEY_LOCAL_MACHINE\Software\Classes</code> .
HKEY_CURRENT_CONFIG	Contains information about the hardware profile that is used by the local computer at system startup.

Accessing registry hives offline (`regedit.exe`)



Accessing registry hives online

- if you only have access to a **disk image**, you must know where the registry hives are located on the disk.
- The majority of these hives are located in the `C:\Windows\System32\Config` directory and are:
 1. **DEFAULT** (mounted on `HKEY_USERS\DEFAULT`)
 2. **SAM** (mounted on `HKEY_LOCAL_MACHINE\SAM`)

3. **SECURITY** (mounted on `HKEY_LOCAL_MACHINE\Security`)
 4. **SOFTWARE** (mounted on `HKEY_LOCAL_MACHINE\Software`)
 5. **SYSTEM** (mounted on `HKEY_LOCAL_MACHINE\System`)
-

Hives containing user information:

- For Windows 7 and above, a user's profile directory is located in `C:\Users\
<username>\` where the hives are:
 1. **NTUSER.DAT** (mounted on `HKEY_CURRENT_USER` when a user logs in)
 - located in the directory `C:\Users\
<username>\`
 2. **USRCLASS.DAT** (mounted on `HKEY_CURRENT_USER\Software\CLASSES`)
 - located in the directory `C:\Users\
<username>\AppData\Local\Microsoft\Windows`
 - Remember that **NTUSER.DAT** and **USRCLASS.DAT** are **hidden files**.
-

The Amcache Hive:

- Windows creates this hive **to save information on programs that were recently run on the system**.
 - This hive is located in `C:\Windows\AppCompat\Programs\Amcache.hve` .
-

Transaction Logs and Backups:

Transaction Logs (Yet to be done)

- The transaction logs can be considered as **the journal of the changelog of the registry hive**.
- Windows often uses transaction logs when writing data to registry hives.
- This means that the transaction logs can often have the latest changes in the registry that haven't made their way to the registry hives themselves.
- The transaction log for each hive is stored as a **.LOG** file in the same directory as the hive itself. It has the same name as the registry hive, but the extension is **.LOG**.
- For example, the transaction log for the SAM hive will be located in `C:\Windows\System32\Config` in the filename **SAM.LOG**.
- Sometimes there can be multiple transaction logs as well. In that case, they will have **.LOG1**, **.LOG2** etc., as their extension.

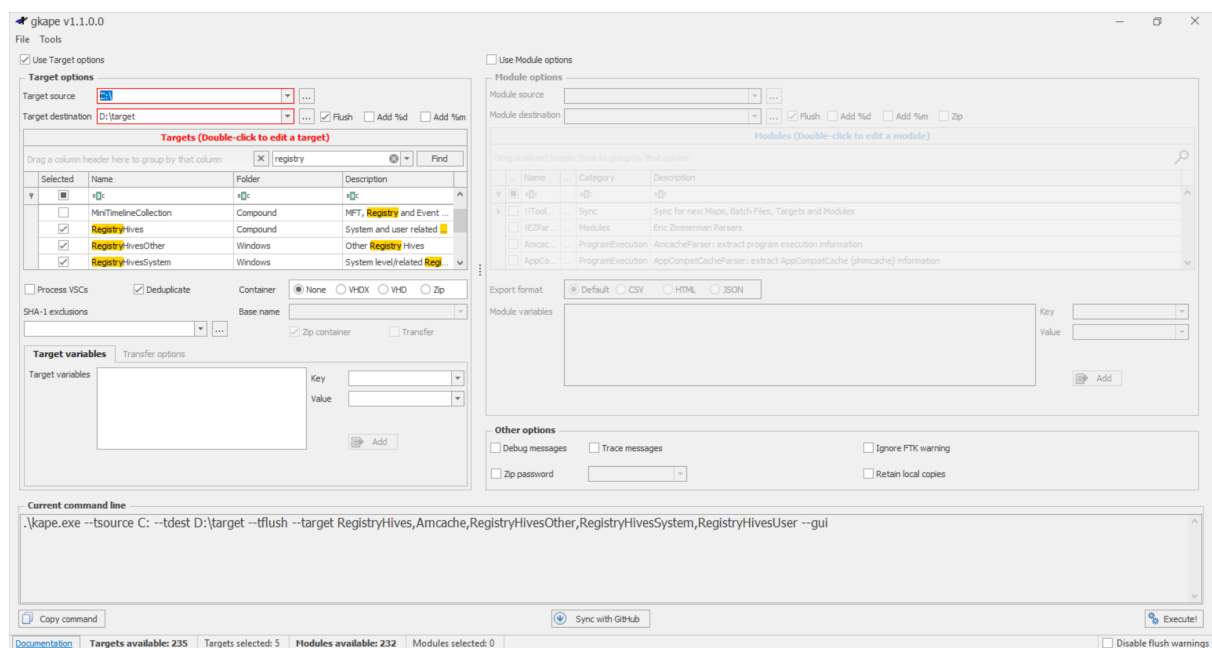
Backups

- Registry backups are the **opposite** of Transaction logs.
- These are the backups of the registry hives located in the `C:\Windows\System32\Config` directory.
- These hives are copied to the `C:\Windows\System32\Config\RegBack` directory **every ten days**.
- It might be an excellent place to look if you suspect that some registry keys might have been deleted/modified recently.

Data Acquisition Tools

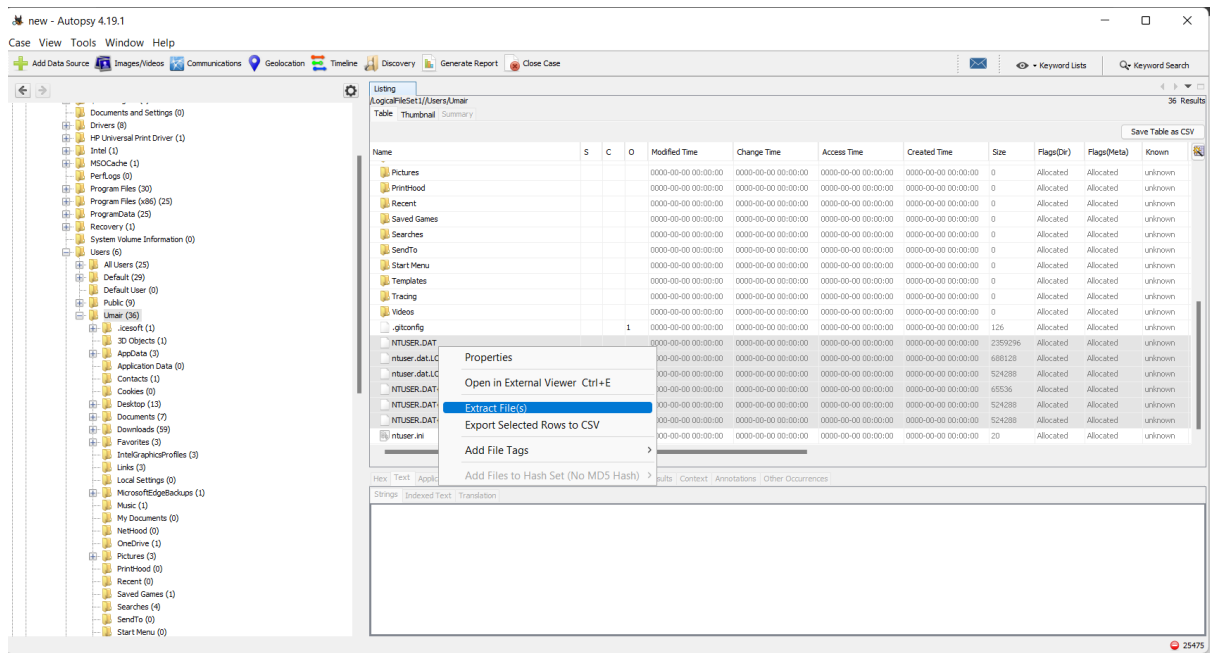
KAPE

KAPE is a live data acquisition and analysis tool which can be used to acquire registry data. It is primarily a command-line tool but also comes with a GUI.



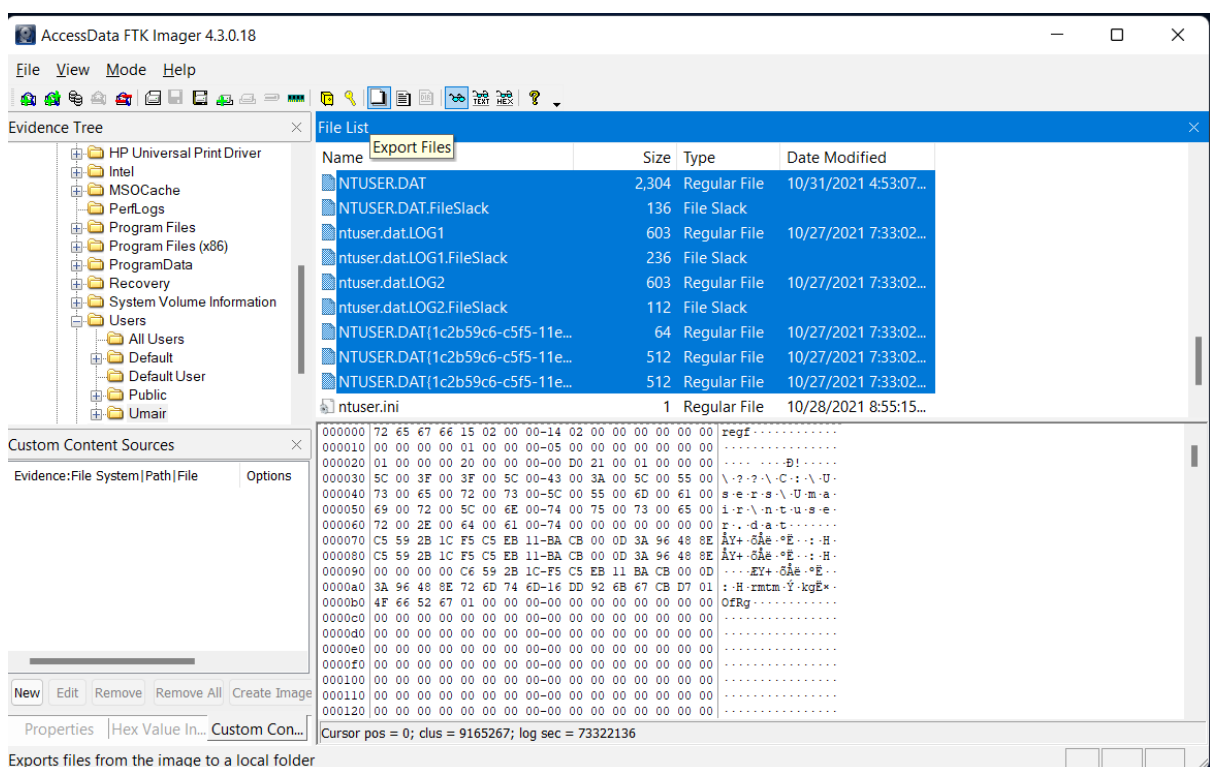
Autopsy

Autopsy gives you the option to acquire data from both live systems or from a disk image. After adding your data source, navigate to the location of the files you want to extract, then right-click and select the Extract File(s) option.



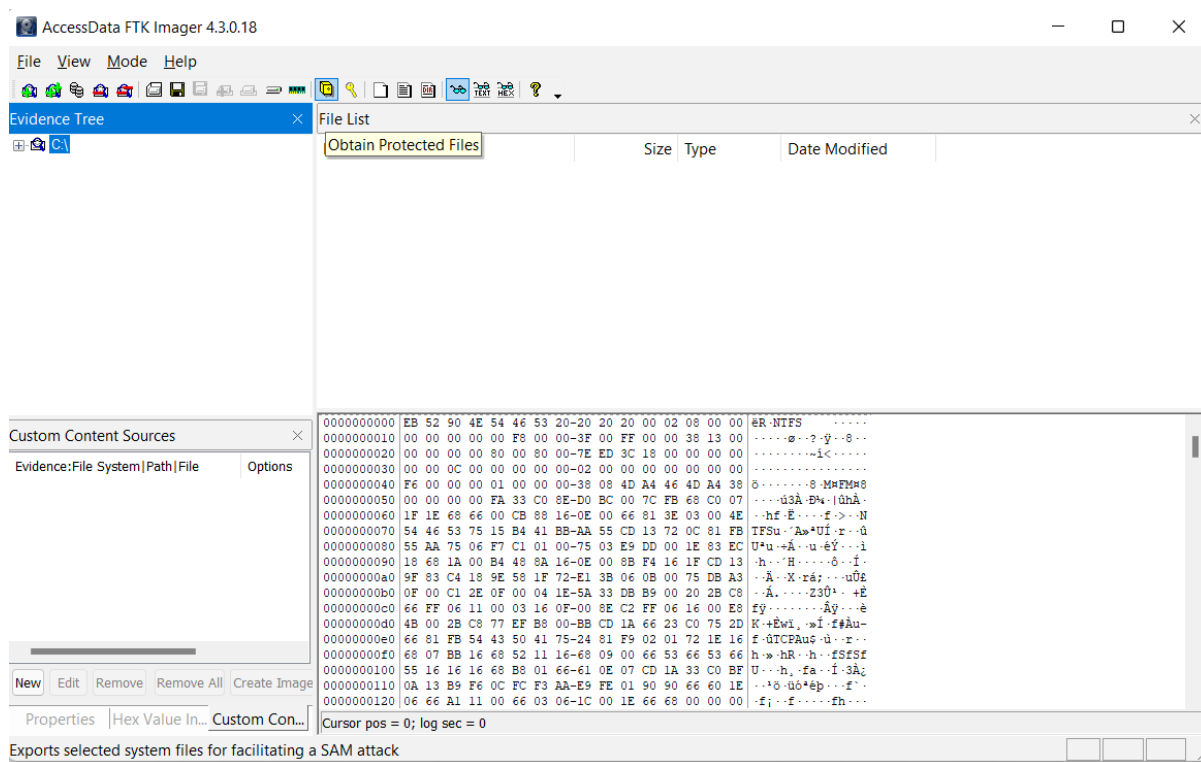
FTK Imager

FTK Imager is similar to Autopsy and allows you to extract files from a disk image or a live system by mounting the said disk image or drive in FTK Imager.



- Another way you can extract Registry files from FTK Imager is through the **Obtain Protected Files** option.

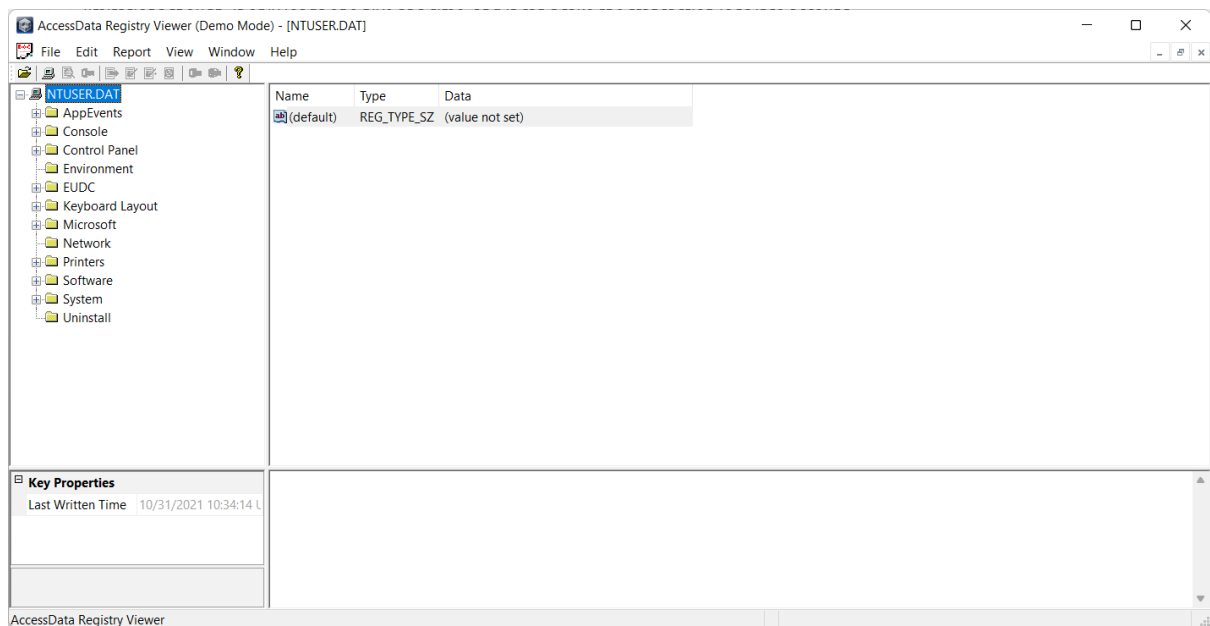
- This option is only available for live systems and is highlighted in the screenshot below. This option allows you to extract all the registry hives to a location of your choosing. However, it will not copy the **Amcache.hve** file



Tools to read/Parse registry

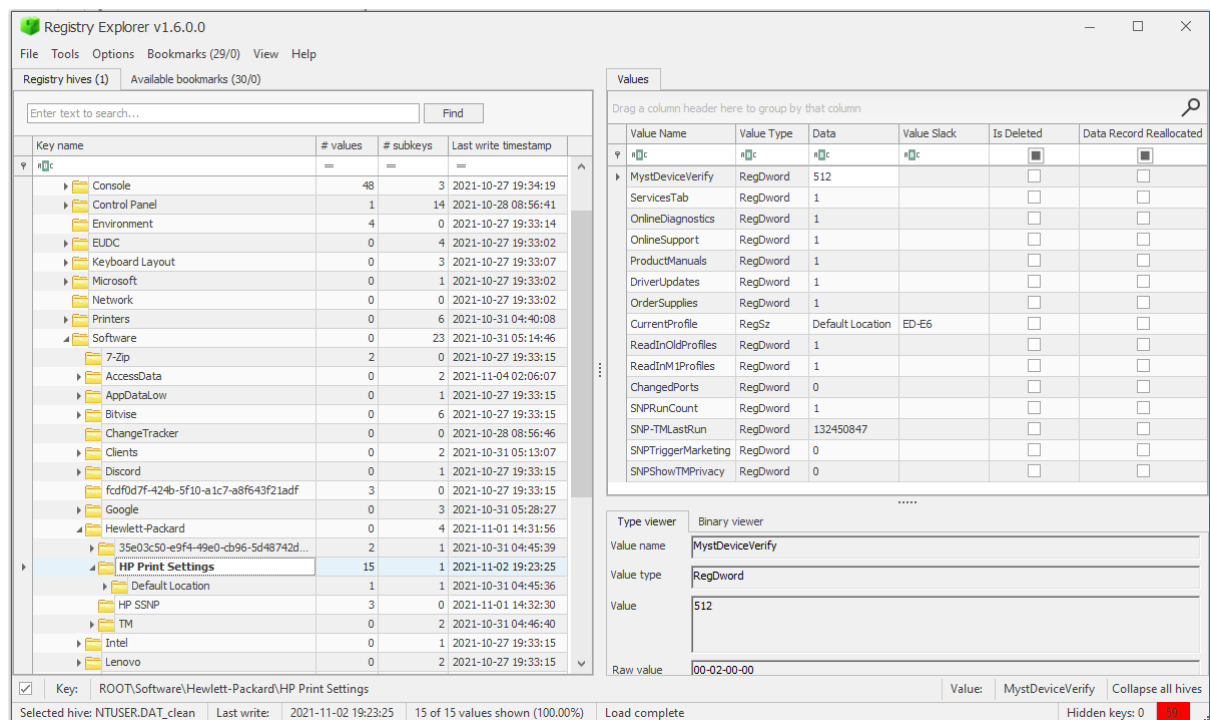
Registry Viewer

- It only loads one hive at a time, and it can't take the transaction logs into account.



Zimmerman's Registry Explorer

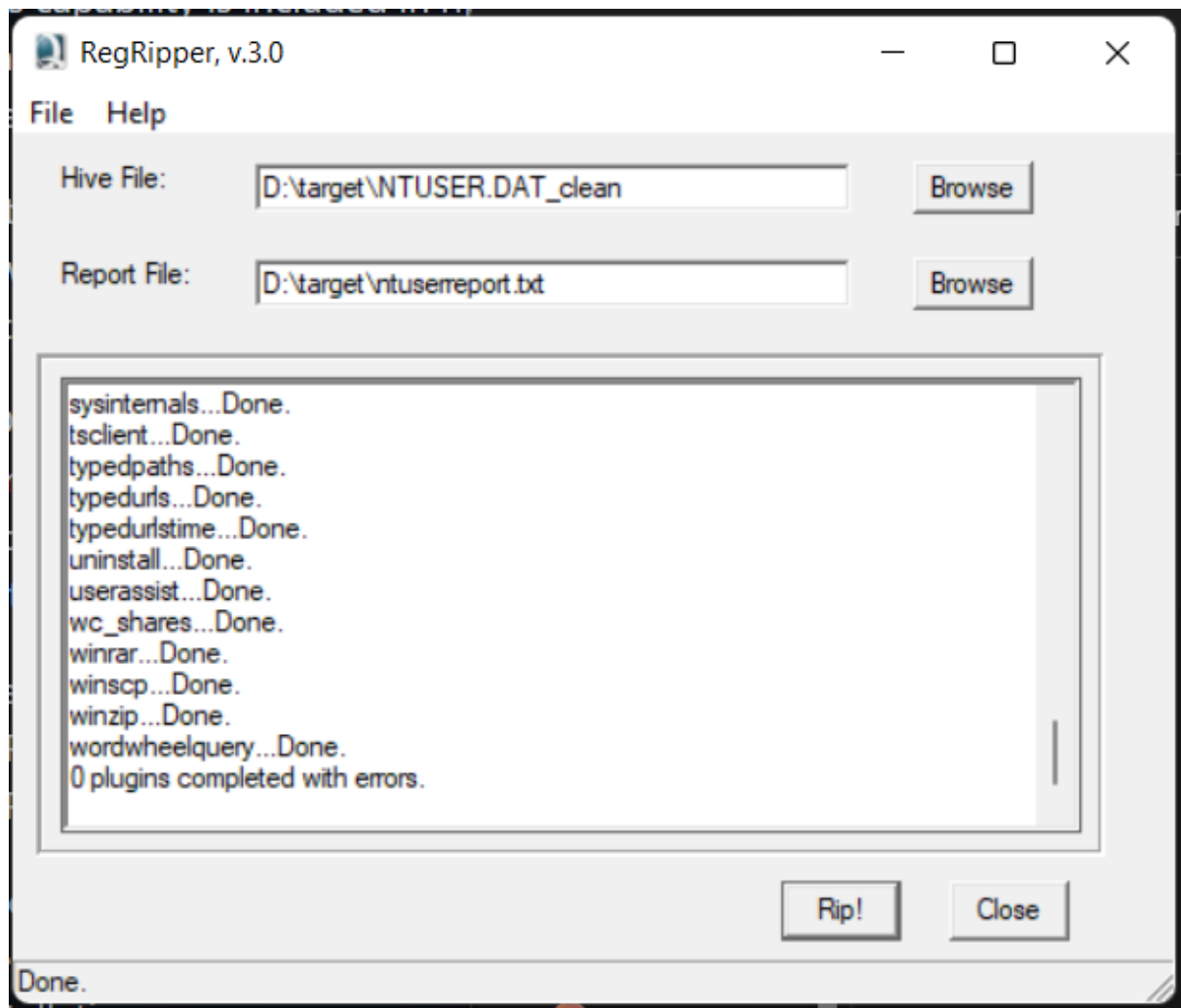
- It can load multiple hives simultaneously and add data from transaction logs into the hive to make a more 'cleaner' hive with more up-to-date data.



RegRipper

- takes a registry hive as input and outputs a report that extracts data from some of the forensically important keys and values in that hive. The output report is in a text file and shows all the results in sequential order.

- One shortcoming of RegRipper is that it does not take the transaction logs into account.



Investigating

System Information and System Accounts

OS Version

- SOFTWARE\Microsoft\Windows NT\CurrentVersion

Key name
▼

▼

- Windows Media Foundation
- Windows Media Player NSS
- Windows Messaging Subsystem
- Windows NT
- CurrentVersion
- Windows Performance Toolkit
- Windows Photo Viewer
- Windows Portable Devices
- Windows Script Host
- Windows Search
- Windows Security Health
- Windows10Upgrader
- WindowsRuntime
- WindowsSelfHost
- WindowsStore
- WindowsUpdate
- Wisp
- WlanSvc
- Wlpasvc
- Wow64
- WSDAPI
- WwanSvc
- XAML
- Monila

Drag a column header here to group by that column
▼

Value Name	Value Type	Data	Value Slack
SystemRoot	RegSz	C:\WINDOWS	00-00-00-00-00-00
BaseBuildRevisionNumber	RegDword	1	
BuildBranch	RegSz	vb_release	00-00-00-00-00-00
BuildGUID	RegSz	ffffffff-ffff-ffff-ffffffff	00-00
BuildLab	RegSz	19041.vb_release.191206-1406	00-00
BuildLabEx	RegSz	19041.1.amd64fre.vb_release.191206-1...	00-00-00-00
CompositionEditionID	RegSz	Enterprise	00-00-00-00-00-05
CurrentBuild	RegSz	19044	
CurrentBuildNumber	RegSz	19044	
CurrentMajorVersionNumber	RegDword	10	
CurrentMinorVersionNumber	RegDword	0	
CurrentType	RegSz	Multiprocessor Free	65-00-64-00-00-00-00-00-00-00-00
CurrentVersion	RegSz	6.3	00-00-00-00
EditionID	RegSz	Professional	00-00
EditionSubManufacturer	RegSz		
EditionSubstring	RegSz		
EditionSubVersion	RegSz		
InstallationType	RegSz	Client	00-00-00-00-00-00
InstallDate	RegDword	1637778211	
ProductName	RegSz	Windows 10 Pro	72-00-70-00-72-00-69-00-73-00-65-00-0...
ReleaseId	RegSz	2009	00-00
SoftwareType	RegSz	System	00-00-00-00-00-00

Autostart Programs (Autoruns)

The following registry keys include information about programs or commands that run when a user logs on.

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Run

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\RunOnce

SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce

SOFTWARE\Microsoft\Windows\CurrentVersion\policies\Explorer\Run

SOFTWARE\Microsoft\Windows\CurrentVersion\Run

Enter text to search... Find

Key name

- RetailDemo
 - Run
 - RunOnce

Drag a column header here to group by that column						
Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated	
SecurityHealth	RegExpandSz	%windir%\system32\Secu...	00-00-00-00	☐	☐	
VMware User Process	RegSz	"C:\Program Files\VMware...	00-00	☐	☐	
VMware VM3DService Process	RegSz	"C:\WINDOWS\system32\...	47-00	☐	☐	

Services info

SYSTEM\CurrentControlSet\Services

- Notice the Value of the Start key in the screenshot below.

Registry hives (7)		Available bookmarks (108/0)		Values			
Enter text to search...		Find		Drag a column header here to group by that column			
Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated		
DependOnService	RegMultiSz	FitMgr	00-00-00-00	☐	☐		
Description	RegSz	@%systemroot%\system...	00-00-00-00	☐	☐		
DisplayName	RegSz	@%systemroot%\system...	00-00-00-00	☐	☐		
ErrorControl	RegDword	1		☐	☐		
Group	RegSz	FSFilter Top	00-00	☐	☐		
ImagePath	RegExpandSz	\SystemRoot\system32\d...	00-00	☐	☐		
Start	RegDword	2		☐	☐		
SupportedFeatures	RegDword	7		☐	☐		
Type	RegDword	2		☐	☐		

In this registry key, if the `start` key is set to 0x02, this means that this service will start at **boot**.

Current control set

- The hives containing the machine's configuration data used for **controlling system startup**.
- Commonly, we will see two Control Sets, **ControlSet001** and **ControlSet002**, in the SYSTEM hive on a machine.
 - In most cases, **ControlSet001** will point to the Control Set that the machine **booted with**
 - `SYSTEM\ControlSet001`
 - ControlSet002 will be the `last known good` configuration.
 - `SYSTEM\ControlSet002`
- Windows creates a volatile Control Set when the machine is **live**, called the CurrentControlSet (`HKLM\SYSTEM\CurrentControlSet`). For getting the most accurate system information, this is the hive that we will refer to.
- We can find out which Control Set is being used as the CurrentControlSet by looking at the following registry value: `SYSTEM\Select\Current`
- Similarly, the `last known good` configuration can be found using the following registry value: `SYSTEM\Select\LastKnownGood`

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated
EnableDHCP	RegDword	1		☐	☐
Domain	RegSz			☐	☐
NameServer	RegSz			☐	☐
DhcpIPAddress	RegSz	192.168.100.58	BA-00-B8-16-0A-00	☐	☐
DhcpSubnetMask	RegSz	255.255.255.0		☐	☐
DhcpServer	RegSz	192.168.100.1	35-00-00-00-65-00-7...	☐	☐
Lease	RegDword	86400		☐	☐
LeaseObtainedTime	RegDword	1637778828		☐	☐
T1	RegDword	1637822028		☐	☐
T2	RegDword	1637854428		☐	☐
LeaseTerminatesTime	RegDword	1637865228		☐	☐
AddressType	RegDword	0		☐	☐
IsServerNapAware	RegDword	0		☐	☐
DhcpConnForceBroadcastFlag	RegDword	0		☐	☐
DhcpNameServer	RegSz	192.168.100.1		☐	☐
DhcpDefaultGateway	RegMultiSz	192.168.100.1	00-00-00-00-00-00	☐	☐
DhcpSubnetMaskOpt	RegMultiSz	255.255.255.0	00-00-00-00-00-00	☐	☐
DhcpInterfaceOptions	RegBinary	FC-00-00-00-00-00-0...	00-00-00-00	☐	☐
DhcpGatewayHardware	RegBinary	C0-A8-64-01-06-00-...	2E-00-30-00-00-00	☐	☐
DhcpGatewayHardwareCount	RegDword	1		☐	☐

The past networks a given machine was connected to can be found in the following locations

SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Signatures\Unmanaged

SOFTWARE\Microsoft\Windows NT\CurrentVersion\NetworkList\Signatures\Managed

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated
ProfileGuid	RegSz	{A3D7C922-7D34-4688...	CA-63-7F-00-CA-99	☐	☐
Description	RegSz	Network 2		☐	☐
Source	RegDword	8		☐	☐
DnsSuffix	RegSz	eu-west-1.compute.int...	F5-48-B1-00-F5-57	☐	☐
FirstNetwork	RegSz	Network 2		☐	☐
DefaultGatewayMac	RegBinary	02-D4-D8-FF-33-74	87-01-C0-51-87-01	☐	☐

- These registry keys contain past networks as well as the last time they were connected.

SAM hive and user information

- The SAM hive contains user account information, login information, and group information.

SAM\Domains\Account\Users

Enter text to search... Find

Key name

C:\Users\THM-4n6\Desktop

ROOT

Console

Control Panel

Environment

ELUDC

Keyboard Layout

Printers

Software

Amazon

Classes

Microsoft

Drag a column header here to group by that column

User Id	Invalid ...	Total L...	Create...	Last Lo...	Last Pa...	Last In...	Expires...	User N...	Full Na...	Passwo...	Groups	Comment	User C...	Home ...	Interne...	Accoun...	Home ...
501	0	0	2021-1...					Guest			Guests	Built-in account for guest access to the computer /domain				☒	☐
503	0	0	2021-1...					DefaultA count			System Managed Accounts Group	A user account managed by the system.				☒	☐
504	0	0	2021-1...		2021-1...			WDAGU8 lityAccou nt				A user account managed and used by the system for Windows Defender Application Guard scenarios				☒	☐
1001	0	19	2021-1...	2021-1...	2021-1...	2021-1...		THM-4n6	count		Administr...					☐	☐

- The information contained here includes the relative identifier (RID) of the user, number of times the user logged in, last login time, last failed login, last password change, password expiry, password policy and password hint, and any groups that the user is a part of.

Usage or knowledge of files/folders

Recent Files

- a list of recently opened files for each user

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs

- located in the directory `C:\Users\<username>\`

Registry hives (7) Available bookmarks (108/0)

Enter text to search... Find

Key name

C:\Users\THM-4n6\Desktop

RecentDocs

Ribbon

RunMRU

SearchPlatform

Shell Folders

Shutdown

StartPage

StuckRects3

TabletMode

Taskband

TypedPaths

User Shell Folders

User Assist

VirtualDesktops

VisualEffects

Wallpapers

Values Recent documents

Drag a column header here to group by that column

Extension	Value Name	Target Name	Lnk Name	Mru Position	Opened On	Extension Last Opened
RecentDocs	7	EZtools	EZtools.lnk		0 2021-12-01 13:00:34	
RecentDocs	6	Settings	Settings.lnk		1	2021-11-30 10:56:23
RecentDocs	5	WallpaperSettings.xml	WallpaperSettings.lnk		2	2021-11-30 10:56:21
RecentDocs	4	System and Security	System and Security.lnk		3	
RecentDocs	3	::{B06C0E4-D293-4F75-8A90-CB0586477EE}	System.lnk		4	
RecentDocs	1	KAPE	KAPE.lnk		5	
RecentDocs	0	Get-KAPEUpdate.ps1	Get-KAPEUpdate.lnk		6	2021-11-24 18:18:48
RecentDocs	2	ChangeLog.txt	ChangeLog.lnk		7	2021-11-24 18:18:48
Folder	2	Settings	Settings.lnk		0 2021-11-30 10:56:23	
Folder	1	System and Security	System and Security.lnk		1	
Folder	0	KAPE	KAPE.lnk		2	
.xml	0	WallpaperSettings.xml	WallpaperSettings.lnk		0 2021-11-30 10:56:21	
.txt	0	ChangeLog.txt	ChangeLog.lnk		0 2021-11-24 18:18:48	
.ps1	0	Get-KAPEUpdate.ps1	Get-KAPEUpdate.lnk		0 2021-11-24 18:18:48	

- if we are looking specifically for the last used **PDF(or any extension)** files, we can look at the following registry key:

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\ .pdf

- located in the directory `C:\Users\<username>\AppData\Local\Microsoft\Windows`

Office Recent Files

- Microsoft Office also maintains a list of recently opened documents.

NTUSER.DAT\Software\Microsoft\Office\<VERSION_Number>

- An example registry key will look like this:

NTUSER.DAT\Software\Microsoft\Office\15.0\Word

- Starting from **Office 365**, Microsoft now ties the location to the user's live ID. In such a scenario, the recent files can be found at the following location.

NTUSER.DAT\Software\Microsoft\Office\VERSION\UserMRU\LiveID_####\FileMRU

ShellBags

- When any user opens a folder, it opens in a specific layout.
- Users can change this layout according to their preferences.
- These layouts can be different for different folders.
- **This information about the Windows 'shell' is stored and can identify the Most Recently Used files and folders.**
- Since this setting is different for each user, it is located in the user hives. We can find this information on the following locations:

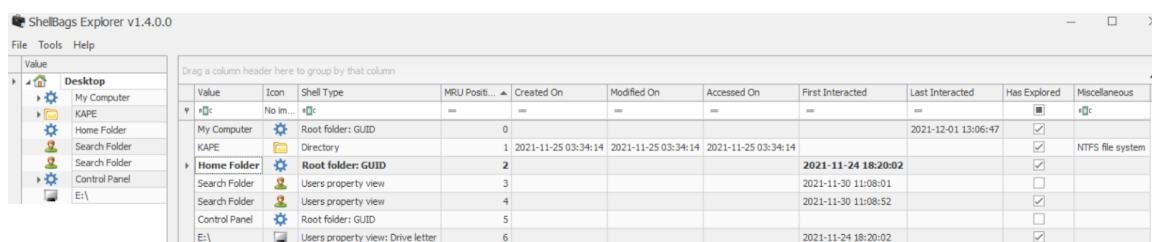
USRCLASS.DAT\Local Settings\Software\Microsoft\Windows\Shell\Bags

USRCLASS.DAT\Local Settings\Software\Microsoft\Windows\Shell\BagMRU

NTUSER.DAT\Software\Microsoft\Windows\Shell\BagMRU

NTUSER.DAT\Software\Microsoft\Windows\Shell\Bags

- Registry Explorer doesn't give us much information about ShellBags.
- However, another tool from Eric Zimmerman's tools called the **ShellBag Explorer** shows us the information in an easy-to-use format. We just have to point to the hive file we have extracted, and it parses the data and shows us the results



Value	Icon	Shell Type	MRU Posit...	Created On	Modified On	Accessed On	First Interacted	Last Interacted	Has Explored	Miscellaneous
My Computer	No im...	Root folder: GUID	0					2021-12-01 13:06:47	☒	
KAPE		Directory	1	2021-11-25 03:34:14	2021-11-25 03:34:14	2021-11-25 03:34:14			☒	NTPS file system
Home Folder		Root folder: GUID	2				2021-11-24 18:20:02		☒	
Search Folder		Users property view	3				2021-11-30 11:08:01		☐	
Search Folder		Users property view	4				2021-11-30 11:08:52		☒	
Control Panel		Root folder: GUID	5						☐	
E:\		Users property view: Drive letter	6				2021-11-24 18:20:02		☒	

Open/Save and LastVisited Dialog MRUs:

- When we open or save a file, a dialog box appears asking us where to save or open that file from.
- It might be noticed that once we open/save a file at a specific location, Windows remembers that location.
- This implies that we can find out recently used files if we get our hands on this information. We can do so by examining the following registry keys

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePidlMRU

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\LastVisitedPidlMRU

Value Name	Mru Position	Executable	Absolute Path	Opened On
0	=	notepad.exe	My Computer\C:\Program Files\Amazon\Ec2ConfigService\Settings	2021-11-30 10:56:19

Windows Explorer Address/Search Bars:

- the paths typed in the Windows Explorer address bar or searches performed

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\TypedPaths

NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\WordWheelQuery

Value Name	Value Type	Data	Value Slack	Is Deleted	Data Record Reallocated
url1	RegSz	C:\	72-00-6F-00-67-00-72-00-61-...	☐	☐
url2	RegSz	C:\Program Files	33-00-32-00-00-00-00-00-00-00	☐	☐
url3	RegSz	C:\Windows\System32	60-53-09-00	☐	☐

Evidence of Execution

UserAssist

- These keys contain information about the **programs launched**, the **time of their launch**, and the **number of times they were executed**.
- However, **programs that were run using the command line can't be found** in the User Assist keys.
- The User Assist key is present in the NTUSER hive, mapped to each user's **GUID**.

NTUSER.DAT\Software\Microsoft\Windows\Currentversion\Explorer\UserAssist\{GUID}\Count

- This data includes **execution path**, **installation**, **execution** and **deletion times**, and **SHA1** hashes of the executed programs.
- This hive is located in the file system at:

`C:\Windows\appcompat\Programs\Amcache.hve`

- Information about the last executed programs can be found at the following location in the hive:

`Amcache.hve\Root\File\{Volume GUID}\`

This is how Registry Explorer parses the AmCache hive:

Timestamp	Path	Name	Product Name	Publisher	Version	SHA1
2021-12-01 12:45:37	c:\program files\windows defender\Microsoft Defender_7.2107.7012.0_x-ww_8b74b3d8b6e33d8ee9.exe	3DViewer.exe	view 3d	microsoft corporation	7.2107.7012.0	3b384600a121040b-fc46-52796773e908996048
2021-12-01 12:55:19	c:\program files\7-zip\7z.exe	7z.exe	7-zip	igor pavlov	19.00	6c7eab88a4435163ae7949c3ae730a9b-0872ae591
2021-12-01 12:55:19	c:\program files\7-zip\7z.exe	7zPM.exe	7-zip	igor pavlov	19.00	e45a108607c8d7398749b8aa72780a3e7a2c5fddca
2021-12-01 12:55:19	c:\program files\7-zip\7z.exe	7zG.exe	7-zip	igor pavlov	19.00	d7226125e47e9404a115d48ebad4903496852508
2021-12-01 13:00:29	c:\program files\google\update\downloader\96.0.4664.45_96.0.4664.45_chrome_installer.exe	96.0.4664.45_chrome_installer.exe		google llc	96.0.4664.45	c2b4b2ae771525b611f54d3e192184a5166e0f
2021-12-01 12:55:49	c:\program files\amazon\amazon-son-agent.exe	amazon-son-agent.exe				e57661919709133855b8c702385545307a30809
2021-12-01 12:57:38	c:\programdata\package cache\77aa2d47-fae7-46c7-8d86-40f213aa49ff\amazon-son-agent.exe	AmazonS3AgentSetup.exe	amazon-son-agent	amazon web services	3.1.338.0	9194f549515d43875e893b49da75ea59682818a
2021-12-01 13:00:20	c:\users\frederik\desktop\amcacheparser.exe	AmcacheParser.exe	amcacheparser	eric stremmer	1.4.0.0	13ab20217d9f43206642d8a22e4545d8c0b3c

BAM/DAM

- **Background Activity Monitor** or **BAM** keeps a tab on the activity of background applications.
- **Desktop Activity Moderator** or **DAM** is a part of Microsoft Windows that optimizes the **power consumption of the device**.
- Both of these are a part of the Modern Standby system in Microsoft Windows.
- In the Windows registry, the following locations contain information related to BAM and DAM.
- This location contains information about **last run programs**, **their full paths**, and **last execution time**.

`SYSTEM\CurrentControlSet\Services\bam\UserSettings\{SID}`

`SYSTEM\CurrentControlSet\Services\dam\UserSettings\{SID}`

Below you can see how **Registry Explorer** parses data from BAM:

Registry hives (3) Available bookmarks (61/0) Values BandName

Enter text to search... Find

Key name

- HKEY_CURRENT_USER\Software\Classes\Local Settings\Software\Microsoft\Windows\CurrentVersion\AppCRD\
 - bam
 - State
 - UserSettings
 - S-1-S-18
 - S-1-S-21-4174496583-1
 - S-1-S-21-417449658**
 - S-1-S-90-0-1
 - S-1-S-90-0-2
 - BasicDisplay
 - BasicRender
 - BattC
 - BaseDrvUserService
 - BaseDrvUserService_7a6b6
 - com2
 - DEESVC
 - keep
 - sFE
 - bandflt
 - BITS
 - BluetoothUserService
 - BluetoothUserService_7a6b6
 - browser

Drag a column header here to group by that column

Program	Execution Time
Microsoft.Windows.ShellExperienceHost_cv5n3h2byewy	2021-11-24 18:02:15
Microsoft.Windows.Cortana_cv5n3h2byewy	2021-11-24 18:02:15
I:\Device\HarddiskVolume2\Windows\explorer.exe	2021-11-24 18:02:15
I:\Device\HarddiskVolume2\Windows\System32\ApplicationFrameHost.exe	2021-11-24 18:02:15
Windows.immersiblecontrolpanel_cv5n3h2byewy	2021-11-24 15:40:31
I:\Device\HarddiskVolume2\Program Files\VMware\VMware Tools\vmtoolsd.exe	2021-11-24 18:02:14
I:\Device\HarddiskVolume2\Windows\System32\cmd.exe	2021-11-25 03:23:14
I:\Device\HarddiskVolume2\Program Files (x86)\Mozilla Firefox\firefox.exe	2021-11-25 03:46:20
I:\Device\HarddiskVolume2\Program Files (x86)\Google\Update\GoogleUpdate.exe	2021-11-25 03:43:40
I:\Device\HarddiskVolume2\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	2021-11-24 17:56:18
I:\Device\HarddiskVolume2\Windows\System32\notepad.exe	2021-11-25 03:42:53
I:\Device\HarddiskVolume2\Users\THM-A6\AppData\Local\Programs\Opera\opera.exe	2021-11-25 04:12:35
I:\Device\HarddiskVolume2\Program Files\Google\Chrome\Application\chrome.exe	2021-11-25 03:43:50
I:\Device\HarddiskVolume2\Windows\System32\mscsc.exe	2021-11-25 04:00:04
I:\Device\HarddiskVolume2\Windows\System32\SystemSettingsAdminFlows.exe	2021-11-25 04:00:54
I:\Device\HarddiskVolume2\Windows\System32\SystemPropertiesComputerName.exe	2021-11-25 04:01:35
I:\Device\HarddiskVolume2\Windows\System32\lun2sc.exe	2021-11-24 17:38:19
I:\Device\HarddiskVolume2\Program Files (x86)\WindowsInstallationAssistant\Windows10UpgraderApp.exe	2021-11-24 18:01:52
I:\Device\HarddiskVolume2\Program Files (x86)\Microsoft Edge\Update\MicrosoftEdgeUpdate.exe	2021-11-24 15:21:35
I:\Device\HarddiskVolume2\Program Files (x86)\Microsoft Edge\Application\msedge.exe	2021-11-24 15:23:43

Total rows: 21 Export

Device identification:

SYSTEM\CurrentControlSet\Enum\USBSTOR

Registry Explorer shows this information in a nice and easy-to-understand way.

First/Last Times:

```
SYSTEM\CurrentControlSet\Enum\USBSTOR\Ven_Prod_Version\USBSerial#\Properties\{83da6326-97a6-4088-9453-a19231573b29}\####
```

Value	Information

0064	First Connection time
0066	Last Connection time
0067	Last removal time

Although we can check this value manually, as we have seen above, Registry Explorer already parses this data and shows us if we select the USBSTOR key.

USB device Volume Name:

The device name of the connected drive can be found at the following location:

`SOFTWARE\Microsoft\Windows Portable Devices\Devices`

Timestamp	Device	Serial Number	Guid	Friendly Name
2021-11-25 07:16:54			{E251921F-4DA2-11EC-A783-001A70DA7110}	USB
2021-11-25 07:16:54			{F529A9D6-4D9C-11EC-A782-001A70DA7110}	New Volume

We can compare the GUID we see here in this registry key and compare it with the Disk ID we see on keys mentioned in device identification to correlate the names with unique devices. Take a look at these two screenshots and answer Question # 3.

Combining all of this information, we can create a fair picture of any USB devices that were connected to the machine we're investigating.

- **reg query HKLM /s /f "C:\TMP\mim.exe sekurlsa::LogonPasswords > C:\TMP\o.txt"**
 - to search for a specific value in the registry